

EDUCATION

- **University of Technology - Vietnam National University Ho Chi Minh City** 01/2026 - Present
M. E. in Engineering Physics (Master Research-intensive Program) Ho Chi Minh City, Vietnam
 - Research: Numerical methods in biomechanics, machine learning, and data science for health.
 - Publications: Q1–Q2 journals and international conferences.
 - Thesis: Development of numerical methods for bone remodeling algorithms with application in biomechanics.
- **University of Technology - Vietnam National University Ho Chi Minh City** 09/2021 - 07/2025
B. E. in Engineering Physics (minor: Biomedical Engineering) (English-taught Program) Ho Chi Minh City, Vietnam
 - Achieved top grades in specialized courses: Fundamentals of Quantum Mechanics and Solid State, Fundamentals of Nuclear Physics and Applications, Engineering Optics and Applications, Microprocessor Applications in Biomedical Engineering, Informatics for Engineering Physics...
 - Teaching assistant: Fundamentals of Biomechanics, Informatics of Engineering Physics, Soft skills.
 - Thesis: Investigation and Simulation of The Mechanical Behavior of Human Skin Layers with The Hyperelastic Model using The Finite Element Method (**Grade: 9.2/10.0**)

WORK EXPERIENCES

- **AI Healthcare Co.,Ltd.** 01/2026 - Present
Vice Director - Researcher Ho Chi Minh City, Vietnam
 - Partner with international universities and companies (Japan, Taiwan, Korea,...) to become the exclusive distributor of medical AI technology in Vietnam
 - Co-organize scientific conferences and seminars on diagnostic imaging and radiotherapy (ISSRT & VART)
- **Branch of Mitalab Co.,Ltd** 09/2024 - 08/2025
Consultant Employee Ho Chi Minh City, Vietnam
 - Designed layout plans for medical equipment placement in hospitals and clinics
 - Optimized diagnostic workflows by over 50%, streamlining both staff operational procedures and equipment utilization.
- **Dong Nai General Hospital** 06/2024 - 08/2024
Engineer Internship Dong Nai Province, Vietnam
 - Assisting the hospital's engineering team in installing and testing medical equipment
 - Participating in discussions with international clients regarding hospital partnerships

RESEARCH EXPERIENCES

- **Lab of Numerical Simulation VIBE Project, HCMUT & Ireland, VNU-HCM** 10/2023 - Present
Student Researcher Ho Chi Minh City, Vietnam
 - Researched: Development of numerical methods for acoustic wave propagation problems and mathematical models of hyperelasticity for biomedical applications (**Participated in 8 research papers at 7 international conferences**) .
 - Mentored 3 undergraduate students through their capstone projects
- **Department of Biomedical Engineering, HCMUT, VNU-HCM** 01/2023 - 06/2025
Student Researcher Ho Chi Minh City, Vietnam
 - Researched machine learning-based analysis and investigation of EEG signals for brain-computer interfaces (**Participated in 2 research papers and 1 coding challenge at 2 international conferences**)
 - Analyzed research data and drafted a manuscript for a peer-reviewed publication

PROJECTS

- **Secure Brain-Computer Interfaces with Homomorphic Encryption: A Case Study on Electroencephalogram Based P300 Classification** 06/2025 - 11/2025
25th Eureka Student Scientific Research Award 2025 (Semi-final Round)
 - On the bigP3BCI dataset with P300 signals, we test time-window and xDAWN features using LDA
 - Paillier HE achieves identical accuracy and AUC as plaintext, ensuring both security with an average accuracy of **85.27%** and an average AUC of **57.21%**
- **MEMORY: A mobile application to support memory improvement for Alzheimer's Patients** 04/2025 - 08/2025
Bach Khoa Innovation 2025 - University of Technology - VNU-HCM (Semi-final Round)
 - The survey was conducted based on the needs of nearly **100** patients, over **200** students, and more than **50** faculty members. The application includes international tests and brain games that were tested at the University.
- **TSHT: Terahertz Sensing and Healing Technology** 02/2024 - 05/2024
IU Startup Demo Day 2024 - International University - VNU-HCM - Grand-final Round
 - Rated as one of the most promising projects in the competition's final round (**Top 6 of 22**). Proposed and developed a Terahertz beam through a crystal
- **Identification and Classification of Multi-label Brain Tumors using MATLAB Software integrated with Graphical User Interface based on Convolutional Neural Network and Deep Learning Algorithms** 11/2023 - 01/2024
BME Innovation 2024 - International University - VNU-HCM - Semi-final Round
 - The brain tumor region was delineated and identified on MRI images, with an F1 result of **98.1%**, and then integrated into a GUI using MATLAB

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT

- [C.1] Nguyen, T. H., Ly, G. H., & **Hoang, D. K. D.** (2026). Leave-One-Subject-Out (LOSO)-Validated Ensemble of Bi-LSTM and Transformer Models for Unusual Activity Detection. *Journal of Physics: Conference Series*, 3180(1), 012008. <https://doi.org/10.1088/1742-6596/3180/1/012008> (SJR: 0.187, ISSN: 1742-6596).
- [C.2] **Hoang, D. K. D.**, & Thi, M. H. N. (2026). A meshfree method based on moving Kriging interpolation for the analysis of acoustic wave propagation problems. *Journal of Physics: Conference Series*, 3180(1), 012060. <https://doi.org/10.1088/1742-6596/3180/1/012060> (SJR: 0.187, ISSN: 1742-6596).
- [C.3] Kha, V. V. H., **Duy, K. D. H.**, Thanh, N. N. (2025). A Moving Kriging Meshfree Approach for Analyzing Large Deformation of Hyperelastic Model. *16th National Conference on Solid Mechanics. Vietnam Journal of Mechanics.* Thai Nguyen Province, Vietnam.
- [C.4] **Duy, K. D. H.**, Trung, N. T., Tich, T. T., Thanh, N. N. (2025). Acoustic Wave Propagation in the Human Ear Canal Using Thermoviscous Boundary Layer Impedance. *16th National Conference on Solid Mechanics.* Thai Nguyen Province, Vietnam.
- [C.5] **Duy, K. D. H.**, Trung, N. T., Tich, T. T., Thanh, N. N. (2025). Nonlinear Simulation of Human Skin under Large Deformations with the Ogden Hyperelastic Model Using a Meshless Method. *The International Symposium on Engineering and Technology 2025 (ISET 2025).* Taichung City, Taiwan.
- [C.6] **Hoang, D. K. D.**, Tran, T. N., Truong, T. T., & Nguyen, T. N. (2025). Simulation of Mechanical Behavior for Human Skin Using Finite Element Method with the Third-Order Ogden Hyperelastic Model. *10th International Conference on the Development of Biomedical Engineering in Vietnam (BME 2024). IFMBE Proceedings, vol 122. Springer, Cham.* 928–939. https://doi.org/10.1007/978-3-031-90194-2_67 (SJR: 0.124, ISBN: 978-3-031-90193-5)
- [C.7] **Duy, K. D. H.**, Trung, N. T., Tich, T. T., Thanh, N. N. (2024). Analysis and Simulation of Microwave Radiation on Human Skin Layers Model using The Finite Element Method. *The International Symposium on Applied Science 2024 (ISAS 2024).* Ho Chi Minh City, Vietnam.
- [C.8] Thanh, L. V., Gia, K. T., Du, A. T., **Duy, K. D. H.**, Thanh, N. N. (2024). Finite Element Analysis of Heat Transfer in Multi-layered Structures. *The 14th OISP Science and Technology Symposium for Students (OSTS14).* Ho Chi Minh City, Vietnam. https://drive.google.com/file/d/1uw_E9coo7foHWxW8No4EyQwdSGdZg8cP/view
- [C.9] Gia, K. T., Thanh, L. V., **Duy, K. D. H.**, Thanh, N. N. (2024). Analysis of Heat Transfer in Aluminum Window Frame. *National Conference on Mechanics - Celebrating the 45th anniversary of the Institute of Mechanics.* Ha Noi City, Vietnam. <https://cohocvietnam.org.vn/media/upload/editor/files/Bia%20%2B%20Muc%20luc%20Tap%20.pdf>

- [C.10] Manh, H. T., & Duy, K. D. H. (2023). Harnessing the Self-Healing Effect of Terahertz Rays: Advancements and Applications. 2023 1st International Conference on Health Science and Technology (ICHST). IEEE Xplore. 1-6.
<https://doi.org/10.1109/ichst59286.2023.10565312> (ISBN: 979-8-3503-1572-1)
- [C.11] Phuc, K. N., Gia, Q. N., Vinh, T. T. H., Thuan, T. L. T., Anh, T. N. N., Duy, K. D. H., Anh, T. T., Quoc, K. L. (2023). Evaluation of Label Noise, Feature Extraction and Classification of Electroencephalogram Signals based on Non-linear Analysis Techniques and Machine Learning Algorithms. *The International Symposium on Applied Science 2023 (ISAS 2023)*. Ho Chi Minh City, Vietnam.

HONORS & AWARDS

- **"Student with 5 good criteria 2025" title - National level** 01/2026
Central Committee of Viet Nam National Union of Students
- **"Advanced youth follow President Ho Chi Minh's wills 2025" title** 05/2025
University of Technology - Vietnam National University Ho Chi Minh City
- **OISP Talented Students 2023-2024** 03/2025
Office for International Study Programs - University of Technology - Vietnam National University Ho Chi Minh City
- **The Potential Project (Grand-final Round) at IU Startup Demo Day 2024** 05/2024
International University - Vietnam National University Ho Chi Minh City
- **Second Prize at the 2022 Engineering Fair, Round 2** 01/2022
Faculty of Applied Science - University of Technology - Vietnam National University Ho Chi Minh City

SKILLS

- **Languages:** MATLAB, Python
- **Tools:** Numpy, Tensorflow, Keras, Pytorch, Scikit-Learn, Matplotlib, Pandas, LaTeX
- **Specialization:** Machine Learning, Deep Learning, AI and Data Science in Health, Numerical Analysis
- **Spoken languages:** Vietnamese (Native), English (Proficient), Japanese (Basic)

LEADERSHIP EXPERIENCE

- **President of OISP Japanese Club** 10/2022 - 10/2023
Ho Chi Minh City University of Technology, Vietnam National University 
- **Mentor of OISP English Club** 10/2022 - 10/2023
Ho Chi Minh City University of Technology, Vietnam National University 
- **President of Ngo Quyen Science and Technology Club** 09/2020 - 09/2021
Ngo Quyen High School, Dong Nai Province 

ACTIVITIES EXPERIENCES

- **Organizing Committee of Ngo Quyen Spring Camp 2023, 2024 and 2025** 01/2023 - 01/2025
Ngo Quyen High School, Dong Nai Province
- **Head of Organizing Committee of OISP Nippon Bunka Matsuri 2023** 24/09/2025
Embassy of Japan in Vietnam and Ho Chi Minh City University of Technology - VNU-HCM 

CERTIFICATIONS

- **Certification of participated in the Winter School on Quantum Computing from November 21-23, 2025 at the University of Technology - Vietnam National University Ho Chi Minh City** 23/11/2025
Vietnam Institute for Advanced Study in Mathematics (VIASM)
- **Certification of Top 40 Bach Khoa Innovation 2025** 19/09/2025
Ho Chi Minh City University of Technology - Vietnam National University
- **Certification of Won third prize in the TOFAS 2024 programming competition - Amateur group** 23/11/2024
Ho Chi Minh City University of Technology - Vietnam National University
- **Certification of completion the Online Continuing Professional Development Program: The 33rd Online Continuing Professional Development Platform for Asian Radiologic Technologist's Societies and The 34th Continuing Professional Development (CPD) Program of Macao Radiological Technologists' Association** 24/01/2024
Macao Radiological Technologists' Association
- **Certification of successfully completed the courses of 2022 Holiday School-Introduction to Biomedical Mechanics from June 20, 2022 to June 27, 2022** 29/06/2022
Department of Biomedical Engineering, National Cheng Kung University

REFERENCES

1. **Assoc. Prof. Dr. Nguyen Thanh Nha** 
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Faculty of Applied Science, Ho Chi Minh City University of Technology, Vietnam National University
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